

Phase 1: Requirement Analysis

Objective

The objective of the Requirement Analysis phase is to identify, analyze, and document all the functional and non-functional requirements of the Community Tool Lending Platform. This ensures that every team member has a clear understanding of the system before the design and implementation phases begin.

Stakeholders

The stakeholders involved in this project are:

- **Borrowers** – Community members who borrow tools.
 - **Owners** – Users who lend their tools.
 - **Administrator** – Manages users, listings, disputes, and the overall platform.
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Functional Requirements

The system shall provide the following functionalities:

User Management

- User registration and login.
- JWT-based authentication.
- Role-based access (Borrower, Owner, Admin).
- Profile management.

Tool Management

- Owners can add, edit, and remove tool listings.
- Upload tool images.
- Specify category, description, condition, availability, and borrowing period.
- Mark tools as unavailable or under maintenance.

Search and Browsing

- Browse available tools.
- Search tools by name or category.
- View tool details and availability.

Reservation Management

- Borrowers can request to borrow a tool.
- Owners can approve or reject requests.
- Prevent double-booking of tools.
- Track reservation status.

Borrowing and Returns

- Record borrowing and return dates.
- Maintain borrowing history.
- Update tool availability after return.

Reviews and Ratings

- Borrowers can rate and review owners or borrowed tools.
- Display reviews to future users.

Notifications

- Notify users about request approvals or rejections.
- Send reminders for upcoming return dates.
- Notify users after successful returns and reviews.

Administration

- Manage users.
 - Remove inappropriate listings.
 - Resolve disputes.
 - View platform statistics and reports.
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Non-Functional Requirements

The system should satisfy the following quality requirements:

- User-friendly interface.
 - Secure authentication using JWT.
 - Reliable data storage using PostgreSQL or MySQL.
 - Fast search and response times.
 - Maintainable and modular architecture.
 - Scalability for future enhancements.
 - Data consistency and integrity.
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Business Rules

- A tool can have only one active reservation during a specific time period.
 - Only the owner of a tool can approve or reject borrowing requests.
 - Only registered users can borrow or list tools.
 - Borrowers can review a tool only after completing a reservation.
 - Administrators have permission to manage all users and listings.
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Assumptions

- All users have internet access.
- Users provide accurate registration information.
- Owners are responsible for maintaining the condition of their tools.
 - The platform focuses on community-based lending rather than commercial rentals.